Preliminary Asset & Load Profile Assessment

Technological Park Project - Almeria

Engineering Department

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Contents

1 Executive Summary & Key Discrepancies

This document outlines the characteristic parameters for the primary electrical assets (A-I) of the proposed Almeria Technological Park. The goal is to provide baseline data for a lumped-parameter model and load-flow analysis.

Before presenting the data, we must highlight several critical discrepancies and assumptions based on the Spanish **Reglamento Electrotécnico de Baja Tensión (REBT)** and standard Medium Voltage (MV) practice:

1. **Workshop C (0.25 MWe Generator) & ITC-BT-40:** The 0.25 MWe generation unit **is not governed by REBT ITC-BT-40**. ITC-BT-40 applies to *Low Voltage* (LV) generating installations. An industrial-scale unit of this size would:
   - Be governed by **High Voltage regulations (RAT)** and the grid code (e.g., RD 337/2014, RD 647/2011).
   - Connect directly to the **20 kV MV bus**, not the LV bus.
   - Require a dedicated **step-up transformer** (e.g., 0.69kV/20kV or 0.4kV/20kV) to inject power into the 20 kV network.
2. **Building Load Scale:** The specified dimensions for the office and museum are of a conventional scale and have been used for load estimation.

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2 Grid Interface & Transformation (A, B, C)

2.1 Element A: Main 20 kV Grid Line (PCC)

Parameters for the Point of Common Coupling (PCC) with the Spanish distribution grid (e.g., e-distribución).

Table 1: Element A: 20 kV Grid Connection Parameters

Parameter	Typical Value / Specification
Nominal Voltage (Un)	20 kV
Frequency (f)	50 Hz
Grid Type	Three-phase, 3-wire
Neutral System	Impedant-grounded or compensated (typical)
Prospective Short-Circuit Power (Pcc)	*Est. 250 MVA* (Requires formal data from DSO)
Regulation	REE Grid Code & DSO specific requirements

2.2 Elements B & C: MV/LV Transformers

Based on the new, more logical load/generation-separated configuration.

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Table 2: Elements B & C: Transformer Parameters

Parameter	Element B (T1: Main Load TX)	Element C (T2: Generator Step-Up)
Loads Served	E, F, G, H, D, I (Auxiliaries)	I (Generator Output)
Transformer Type	Oil-filled (or dry-type)	Oil-filled (or dry-type)
Nominal Rating (S)	**Est. 2,500 kVA (2.5 MVA)**	**Est. 400 kVA (0.4 MVA)**
	(Sized for all park loads)	*(Sized for 0.25MWe generator export)*
Voltage Ratio	20 kV / 400V-230V (Step-Down)	**Est. 0.4kV / 20kV (Step-Up)**
Vector Group	Dyn11 (Standard Distribution)	YNd1 (Typical GSU)
Short-circuit V (uk)	$\sim$ 6%	$\sim$ 4-5%
Regulation	REBT ITC-BT-19, RAT	**RAT & Grid Code (RD 647)**

3 Consumer Load Profiles (D, E, F, G, H)

3.1 Element D: Perimeter Streetlamps

Table 3: Element D: Streetlamp Load

Parameter	Value / Specification
Quantity	18 units
Load Type	LED (Assumed for modern park)
Est. Power per Unit	150 W
Total Peak Load (P)	$18 \times 150W =$ **2.7 kW**
Power Factor ($\cos\phi$)	> 0.95 (Characteristic of LED with drivers)
Regulation	REBT ITC-BT-09

3.2 Elements E, F, G, H: Building Loads

Lumped-parameter estimates for the main consumer buses.

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4 Generation Asset (I)

4.1 Element I: 0.25 MWe Research Power Plant

As noted, this is an **MV Generation Asset**, not an LV installation under ITC-BT-40. It will act as a source bus, injecting power into the 20 kV line. It will also be a small consumer for its own auxiliary services.

Table 4: Elements E-H: Consumer Load Profiles

Ref	**Description**	**Key Internal Loads**	**Est. Peak Load (P)**	**Est. Power Factor (cos ϕ)**
E	Office Building (3,600 m^2)	HVAC, IT/Servers, Lighting (LED), Elevators	216 kW (at 60 W/m^2)	$\sim$ 0.9 (Corrected)
F	Public Museum (21,600 m^2)	Critical HVAC (24/7), Specialty Lighting, Security, Exhibits	864 kW (at 40 W/m^2)	$\sim$ 0.9 (Corrected)
G	Workshop A (Machining) (8,000 m^2)	CNC machines, VFDs, large motors, compressors (Inductive)	500 kW	**0.7 - 0.8** (Uncorrected)
H	Workshop B (Chemical) (8,000 m^2)	Heaters, ovens, resistive furnaces, lab equipment (Resistive)	400 kW	**> 0.98**

Table 5: Element I: 0.25 MWe Generation Plant

Parameter	**Value / Specification**
Type	Research Generator (Small Modular Nuclear) (in 8,000 m^2 facility)
Nominal Thermal Power (Pth)	**1.0 MWth**
Nominal Active Power (P)	**0.25 MW (250 kWe)**
Apparent Power (S)	$\sim$0.3 MVA (Assuming cos ϕ = 0.85)
Generation Voltage	Est. 400V (to match LV bus)
Key Required Element	**Dedicated Step-Up TX (Element C)**
Connection Point	20 kV MV Bus (via Element C)
Governing Regulation	**RD 337/2014 (RAT)**, **RD 647/2011** (Not REBT ITC-BT-40)
Operational Mode	Grid-injection, self-consumption
Key Parameters	Active/Reactive Power capability, P-Q curve, fault-ride-throug